2013 F=ma Exam: Problem 18

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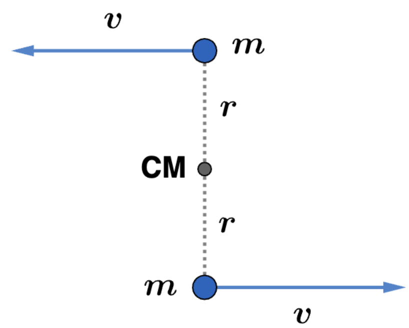


Since there are no external forces (and torques), we have conservation of linear momentum and angular momentum. The initial linear momentum is

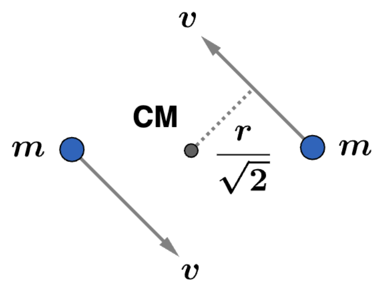
$$p_i = mvr - mvr = 0$$

Since $p_{\text{tot}} = M_{\text{tot}} v_{\text{CM}}$, this means $v_{\text{CM}} = 0$ so the CM stays at rest at its initial location $(1, 0)$. Choosing $(1, 0)$ as our axis of rotation, the initial angular momentum is

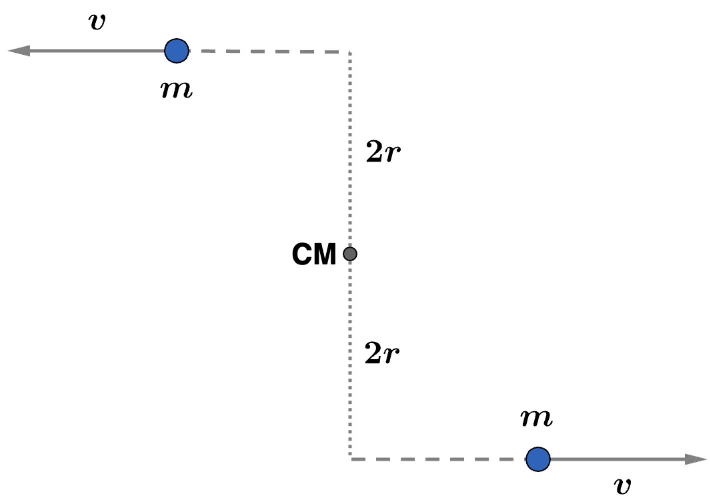
$$L_i = 2mvr = 2(1\,\text{kg})(1\,\text{m/s})(1\,\text{m}) = 2\,\text{kg}\,\text{m}^2/\text{s}$$

going counterclockwise. We now go through the possible choices:

- A) Not correct because the CM moved to $(0, 0)$.
- B) Not correct because the linear momentum is nonzero.
- C) Not correct because the angular momentum decreased (since the moment arm decreased by a factor of $\sqrt{2}$).



- D) Not correct because the angular momentum increased (since the moment arm increased by a factor of 2).



- E) Correct since linear momentum and angular momentum are conserved and the CM stayed at rest.

Thus, the answer is $\boxed{\text{E}}$.